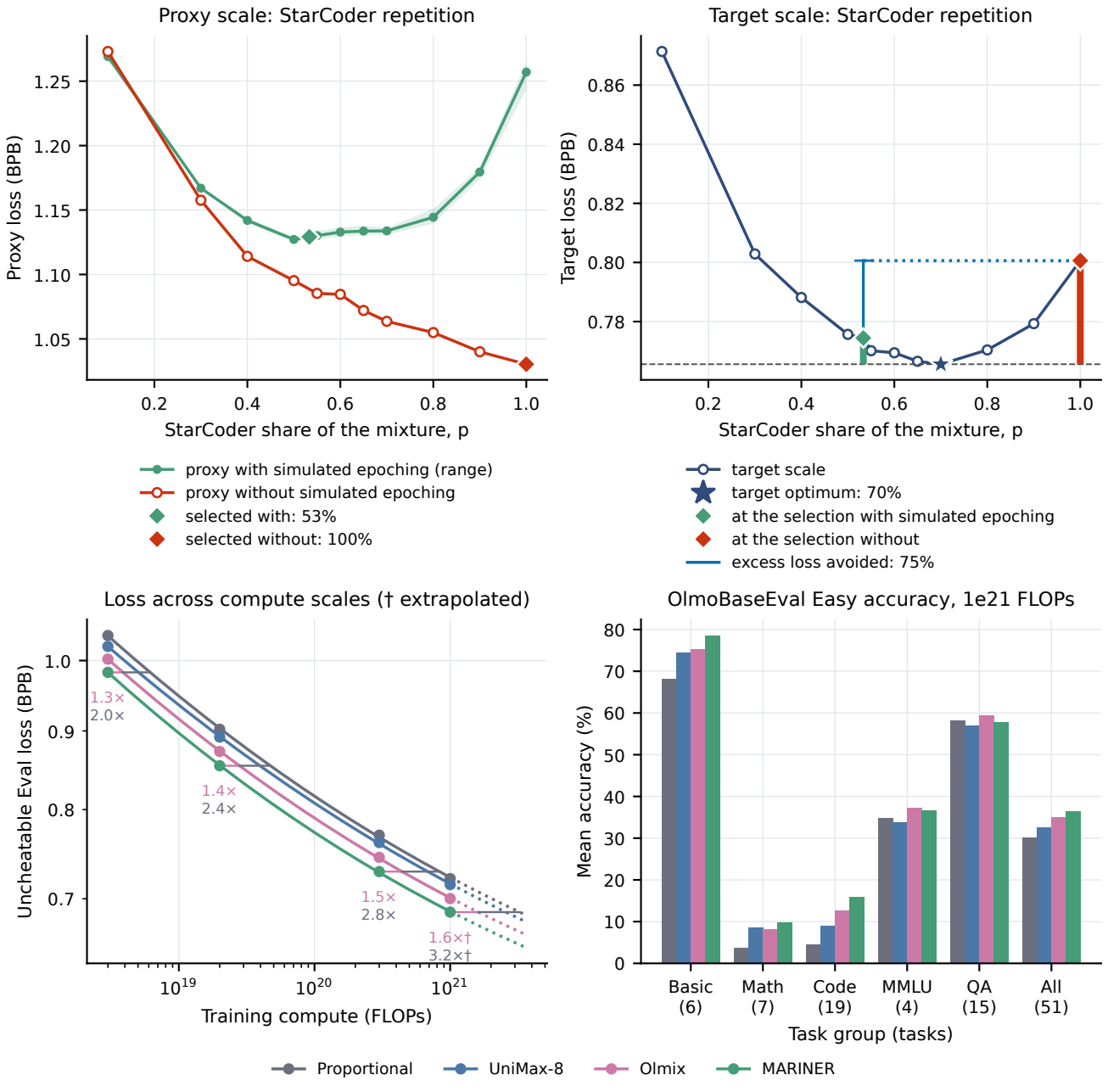
Proxy scale: StarCoder repetition
Proxy loss (BPB)
1.25
1.20
1.15
1.10
1.05
0.2
0.4
0.6
0.8
1.0
StarCoder share of the mixture, p
proxy with simulated epoching (range)
proxy without simulated epoching
selected with: 53%
selected without: 100%
Target scale: StarCoder repetition
Target loss (BPB)
0.86
0.84
0.82
0.80
0.78
0.2
0.4
0.6
0.8
1.0
StarCoder share of the mixture, p
target scale
target optimum: 70%
at the selection with simulated epoching
at the selection without
excess loss avoided: 75%
Loss across compute scales († extrapolated)
Uncheatable Eval loss (BPB)
1.0
0.9
0.8
0.7
1.3×
2.0×
1.4×
2.4×
1.5×
2.8×
1.6×†
3.2×†
10^{19}
10^{20}
10^{21}
Training compute (FLOPs)
OlmoBaseEval Easy accuracy, 1e21 FLOPs
Mean accuracy (%)
80
70
60
50
40
30
20
10
0
Basic (6)
Math (7)
Code (19)
MMLU (4)
QA (15)
All (51)
Task group (tasks)
Proportional
UniMax-8
Olmix
MARINER